Monetary Policy Transmission and Building Land Scarcity: A Panel Local Projections Analysis of Regional Austrian Housing Markets

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Abstract

Acknowledgements

Contents

1. Introduction **2**
1.1. Motivation . . . 2
1.2. Research Gap . . . 2
1.3. Research Question . . . 2
1.4. Contribution . . . 2
1.5. Thesis Outline . . . 2

2. Institutional and Economic Background **2**
2.1. ECB Monetary Policy Context . . . 2
2.2. The Austrian Real Estate Market, 2017–2025 . . . 2
2.3. Regional Housing Market Heterogeneity . . . 2

3. Literature Review: Theory and Hypotheses **2**
3.1. Monetary Policy and Housing Prices . . . 2
3.2. Housing Supply Constraints and Monetary Policy Transmission . . . 2
3.3. Austrian Housing and Land Price Evidence . . . 2
3.4. Remaining Gap and Hypotheses . . . 2

4. Data and Descriptive Evidence **2**
4.1. Real Estate Price Data . . . 3
4.2. Monetary Policy Shocks . . . 7
4.3. Supply-Constraint Measure . . . 9
4.4. Control Variables . . . 10
4.5. Panel Construction and Preliminary Diagnostics . . . 15

5. Empirical Strategy **15**
5.1. Panel Local Projections . . . 16
5.2. Baseline Model and Identification . . . 17
5.3. Outcome Transformation . . . 18
5.4. Interpretation of the Coefficient . . . 19
5.5. Fixed Effects . . . 20
5.6. Inference and Identification Assumptions . . . 22
5.7. Alternative Specifications and Robustness . . . 24

6. Empirical Results **26**
6.1. Baseline Results 27
6.2. Alternative Specifications 29
6.3. Robustness Checks 33

7. Discussion **36**
7.1. Interpretation of Findings 36
7.2. Relation to Literature 36
7.3. Economic Relevance 36
7.4. Policy Implications 36
7.5. Limitations 36

8. Conclusion **36**

References **37**

A. Appendix **39**

1. Introduction

1.1. Motivation

1.2. Research Gap

1.3. Research Question

1.4. Contribution

1.5. Thesis Outline

2. Institutional and Economic Background

2.1. ECB Monetary Policy Context

2.2. The Austrian Real Estate Market, 2017–2025

2.3. Regional Housing Market Heterogeneity

3. Literature Review: Theory and Hypotheses

3.1. Monetary Policy and Housing Prices

3.2. Housing Supply Constraints and Monetary Policy Transmission

3.3. Austrian Housing and Land Price Evidence

3.4. Remaining Gap and Hypotheses

4. Data and Descriptive Evidence

The empirical analysis rests on a balanced quarterly panel of Austrian districts covering the period Q1 2017 to Q4 2025. The political district is the spatial unit. The panel

therefore comprises 116 districts observed over 36 quarters, which yields 4 176 district-quarter observations. Vienna is retained at the level of its individual municipal districts, so that the Viennese housing market enters the analysis at the same granularity as the remaining political districts rather than as a single aggregate. The following subsections describe the outcome variable, the monetary policy shock, the supply-constraint measure, and the control variables in turn. Further, the subsections document descriptive evidence that motivates the following empirical analysis, before the closing subsection documents how these components are combined into the estimation panel along with preliminary diagnostics of the data.

4.1. Real Estate Price Data

Residential real estate prices are provided by "DataScience Service GmbH" (2026), which compiles district-level price measures from transaction data. The data distinguish three property categories, average prices for single-family houses (HOUSE), average prices for condominium apartments (APT), and building land prices, each expressed in € per m². These data are based on land-register transaction records ("Grundbuchauszüge"). Building land is not treated as a housing outcome and is therefore excluded from the price series used in estimation. Over the full sample the price data are complete, with an observation for every district in every quarter and no missing values in any of the three categories.

The primary outcome is a district-level composite price constructed as a fixed-weight average of the house and apartment series. For district i in quarter t, the composite price level is defined as

$$p_{i,t}^{\text{comp}} = w_i^{\text{house}} \, \text{HOUSE}_{i,t} + w_i^{\text{apt}} \, \text{APT}_{i,t}, \tag{4.1}$$

where $\text{HOUSE}_{i,t}$ and $\text{APT}_{i,t}$ are the average house and apartment prices in € per m², and the weights w_i^{house} and w_i^{apt} are held constant over the sample and are derived from the district housing stock, as described in the next paragraph. The underlying price data are in nominal terms. No district-level deflator is available at quarterly frequency, but this poses no difficulty for the analysis, since the quarter fixed effects used in the empirical strategy absorb national inflation along with every other development common

to all districts in a quarter. The composite is formed at the level of prices and with fixed weights, so that cross-district property composition is held constant and only the underlying prices vary over time. This construction also motivates examining the house and apartment series separately as an alternative specification, and it provides the price series that is later transformed into the dependent variable.

The composite weights are based on the building stock recorded in the 2011 Statistik Austria building and housing census, which reports the number of dwellings by the number of dwellings within the building on the district-level (Statistik Austria, 2026d). Let D_i^1, D_i^2, $D_i^{3\text{–}5}$, $D_i^{6\text{–}10}$, $D_i^{11\text{–}20}$, and D_i^{21+} denote the 2011 dwelling counts in district i across the six building-size classes. Dwellings in single-dwelling and two-dwelling buildings are assigned to the house category, the latter to capture semi-detached and duplex dwellings, while dwellings in buildings with three or more dwellings are assigned to the apartment category. The house weight is then the house-category share of the district dwelling stock,

$$w_i^{\text{house}} = \frac{D_i^1 + D_i^2}{D_i^1 + D_i^2 + D_i^{3\text{–}5} + D_i^{6\text{–}10} + D_i^{11\text{–}20} + D_i^{21+}}, \qquad w_i^{\text{apt}} = 1 - w_i^{\text{house}}. \tag{4.2}$$

The weights are fixed at their 2011 values so that they predate the estimation window and remain exogenous to price developments during the sample, following the same logic that governs the supply-constraint measure below.

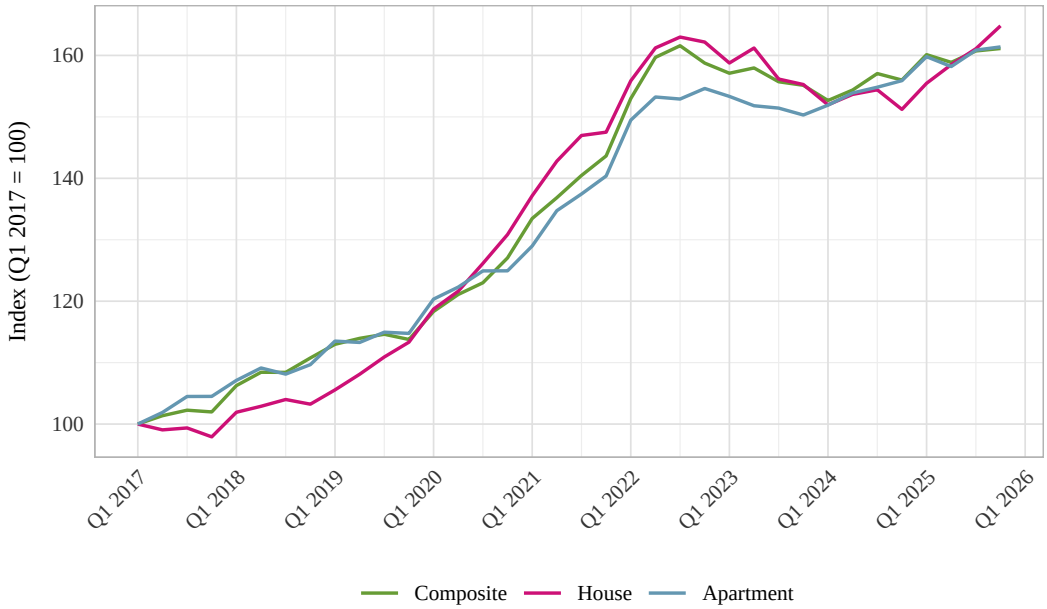


Figure 1: District price indices by property type
Unweighted district medians, Q1 2017 = 100.

Averaged over all district-quarter observations, that is, across the 116 districts and 36 quarters of the panel, the composite price is roughly € 3 260 per m², with a standard deviation across those district-quarter observations of about € 1 590 and a range that spans from under € 900 to over € 13 000 per m². This wide dispersion is the first indication of substantial regional heterogeneity in Austrian housing markets. The house and apartment series differ markedly in level, with average house prices around € 4 200 per m² exceeding average apartment prices of about € 2 870 per m². Both series are right-skewed, so that a small number of high-price districts sit well above the median. The evidence presented is descriptive and is not intended to support causal claims, since the differences reported can reflect urbanity, dwelling composition, income, and other district characteristics rather than the transmission of monetary policy. The purpose is to establish that the data contain economically meaningful regional heterogeneity in price levels, price dynamics, and supply conditions, and to surface the features that shape the specification and interpretation of the local projections analysis.

The price dispersion is closely related to supply conditions. Grouping districts by the supply-constraint classification (see Section 4.3), the mean composite price, again pooled over all district-quarter observations, is roughly € 4 410 per m² in inelastic districts

against about € 2 480 in elastic districts, so that the more supply-constrained districts operate in a distinctly higher price regime. As emphasized above, this level difference is not evidence of differential monetary transmission, since inelastic districts are also more urban and more apartment-heavy. It does, however, confirm that the supply classification separates districts that differ systematically in their housing markets, which is a precondition for the interaction analysis that follows.

More important for the empirical strategy is the time-series behavior of price growth. Quarterly composite price growth averages about 1.3% over the sample, with elastic districts growing marginally faster on average than inelastic districts. Pooling the within-district changes, the autocorrelation (ACF) of quarterly composite growth is positive and persistent, at about 0.23 at the first lag and remaining close to 0.20 up to the fourth lag. Positive autocorrelation means that a quarter of above-average price growth tends to be followed by further above-average growth, so that shocks to prices are not reversed immediately but propagate over several quarters. Price momentum is a well-documented feature of housing markets (Case & Shiller, 1988; Guren, 2018), and reflects the slow diffusion of information, search frictions, and the staggered nature of transactions in illiquid property markets.

The presence of price momentum has a direct methodological implication. If lagged price changes predict current growth and are also correlated with the timing of monetary policy surprises, omitting them would confound the estimated response with the ongoing momentum of the local market. Following Aastveit and Anundsen (2018), who include four lags of house-price changes in their panel local projections precisely to absorb this momentum, the control vector therefore includes four lagged quarterly changes of the composite price. The lag profile in Figure 2 supports this choice. The partial autocorrelation (pACF), which measures the additional predictive content of each further lag once the earlier lags are accounted for, is positive and significant up to the fourth lag and then sharply falls at the fifth, so four lags capture the independent momentum content without adding redundant terms. The negative correlations at longer lags are consistent with a differenced series and do not reflect the positive momentum the controls are meant to absorb. The same persistence, combined with the overlapping cumulative outcomes of the local projections, is also a source of serial correlation in the residuals, which is why the treatment of standard errors requires particular care in the empirical strategy.

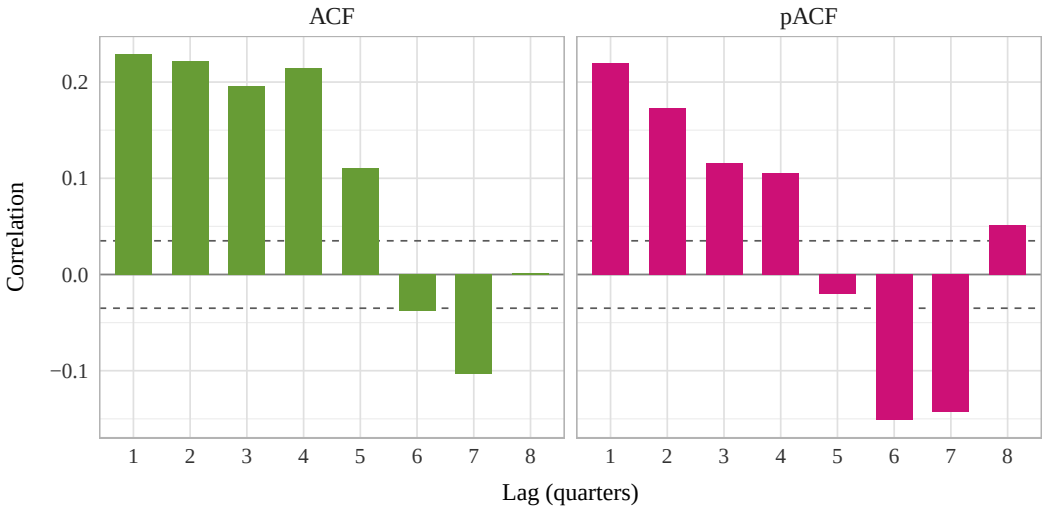


Figure 2: Autocorrelation and partial autocorrelation of quarterly composite price growth
Pooled within-district estimates by lag. Dashed lines mark the approximate white-noise significance band, $\pm 1.96/\sqrt{n}$.

4.2. Monetary Policy Shocks

The monetary policy shock is constructed from the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019), which records high-frequency changes in a range of asset prices within tight windows around ECB policy announcements. The chosen instrument is the surprise in the twenty-year overnight index swap rate (OIS 20Y). This maturity is selected because residential real estate prices depend on expected long-term financing conditions rather than on the current policy rate. Housing is a long-lived asset, and its value can be approximated by the discounted stream of expected future service flows (rents or imputed rent), so a change in the long-term discount rate has a larger valuation effect than a change in short rates (He & La Cava, 2020), a channel that operates through housing wealth in the euro area (Di Casola & Grothe, 2026). Consistent with this, the transmission of ECB policy, and of its asset purchase programmes in particular, works to a large extent through the long end of the yield curve (Eser et al., 2019), so that surprises at longer maturities are the component of monetary policy most relevant for housing valuations. The twenty-year and thirty-year German government bond yield surprises are retained only as alternative instruments

for robustness.

The event-level surprises are aggregated to the quarterly frequency of the panel by summing all surprises within a quarter into a net quarterly shock. Alongside the net measure, the sum of positive surprises, the sum of negative surprises, and the sum of absolute surprises are recorded for each quarter, which supports the signed decomposition used to examine asymmetric responses. The sign convention follows the interpretation of the underlying instrument, a negative OIS 20Y surprise denotes a downward revision of long-term rate expectations and therefore a monetary easing surprise. This convention matters for the interpretation of the estimated coefficients and is carried through consistently in the empirical analysis. Over the sample, the mean net quarterly surprise is negative, at roughly -0.59, consistent with the prevailing monetary easing stance during much of the period, and the largest absolute net surprises occur in Q3 2019 and around the onset of the tightening cycle in 2022 and 2023.



Figure 3: Quarterly monetary policy shock components
Easing surprises denoted as negative, tightening surprises denoted as positive; points = net quarterly sum

A preliminary diagnostic examines how far the quarterly net sum conceals offsetting surprises within a quarter. The cancellation ratio, defined as the net shock (net sum of

shocks in a quarter) divided by the absolute sum of surprises in a quarter, has a mean of about 0.69, which indicates that easing and tightening surprises frequently occur within the same quarter and partly offset in the net measure. The ratio is a diagnostic rather than an additional regressor, and the extent of within-quarter cancellation motivates an asymmetric specification that separates the easing and tightening components of the shock, which the empirical strategy adopts as an alternative specification.

Both easing and tightening quarters are well represented, with an easing surprise component present in 31 of the 36 quarters and a tightening surprise component in 26, and both signs occurring within the same quarter in 21 quarters. This overlap is the source of the within-quarter cancellation documented above, and the presence of both directions of surprise is what makes the sample suitable for examining whether price responses are asymmetric across monetary policy phases. Because the shock is common to all districts within a quarter, its aggregate effect is absorbed by the time fixed effects, and the descriptive variation reported here bears on the identifying interaction only through its differential incidence across the supply groups.

4.3. Supply-Constraint Measure

Regional differences in housing supply are captured through the district building land reserve recorded in the ÖROK monitoring of building land reserves (ÖROK Monitoring of Building Land Reserves, 2025), which measures the share of vacant developable land in total designated building land. A lower reserve indicates less room for additional construction and therefore a more supply-inelastic district. To predate the estimation window, the 2014 level of the reserve is used throughout, so that the classification cannot respond to price developments during the sample.

Districts are classified as supply-inelastic using the twenty-fifth percentile of the reserve distribution outside Vienna, a threshold of around 22%, which is then applied to all districts. This procedure classifies 47 districts as inelastic and 69 as elastic. Of the inelastic districts, 24 lie outside Vienna and 23 are Viennese municipal districts. The measure is a proxy for supply conditions rather than a directly estimated supply elasticity, and it is strongly structured along geographic and compositional lines. In particular, the reserve measure correlates at about 0.91 with the fixed house weight, so that inelastic districts tend also to be apartment-heavy. The interpretation of this entanglement is developed

below, alongside the correlation evidence. District fixed effects absorb time-invariant level differences of this kind, but the entanglement of supply constraints with geography and property composition remains a concern for interpretation. For this reason the analysis later reports property-specific outcomes and a continuous reserve measure as alternative specifications, and re-estimates the model on the sample excluding Vienna as a robustness check.

The building land reserve varies considerably across districts, with a median of about 24% and values ranging from close to zero to above 45%. The supply classification is strongly clustered by federal state. All Viennese districts and all Salzburg districts fall into the inelastic group, while Burgenland contains no inelastic districts, and the median reserve in Vienna, at under 5%, is far below that of the predominantly rural states. This geographic concentration is economically intuitive, since the most supply-constrained districts are the dense urban and Alpine areas where developable land is genuinely scarce, but it also implies that the supply measure is entangled with region and with property composition.

This entanglement carries two implications that are taken up in the empirical analysis. First, district fixed effects remove time-invariant differences in price levels and composition, but they do not by themselves rule out that districts differing in these correlated characteristics also respond differently to a common shock for reasons other than supply elasticity. Second, because the interaction with the shock is by construction non-zero only for inelastic districts, and because all Viennese districts are inelastic, the identifying variation is concentrated in a structurally distinctive set of markets. These considerations motivate the alternative specifications reported alongside the baseline, the continuous reserve measure and the property-specific outcomes, as well as the Vienna-excluded robustness check, and they caution against reading the baseline interaction as a clean measure of supply elasticity alone.

4.4. Control Variables

The control set comprises district-level population, the unemployment rate, and net income per capita, each aligned to the quarterly frequency of the panel. Population is available quarterly from the central residence register and enters directly (Statistik Austria, 2026b). Unemployment is available annually at the district level and is carried

across the four quarters of each year (Statistik Austria, 2026a). Net income per capita is constructed by dividing annual district net income totals (Statistik Austria, 2026e) by the corresponding population, so that within-year variation in income per capita partly reflects population movements. This construction creates a mechanical correlation between income growth and population growth. Since these series are annual in origin, the quarterly panel repeats the annual figures within each year, and this is reflected in the way the control lags behave.

Several series require construction at the tail of the sample, where source data are not yet available. Annual net income is observed through 2022 and is extrapolated for 2023 to 2025 using the nominal Tariflohnindex (OeNB - Österreichische Nationalbank, 2026), so that all districts share a common wage-index growth path over the extrapolated years. District unemployment is observed through 2024 and carried forward to 2025, and population is carried forward from the first quarter of 2025 to the remaining quarters of that year. These constructed values are complete observations, but they do not represent newly observed district information. Their share is largest at the end of the sample and declines at longer horizons as later quarters drop out of the estimation window. To ensure that these choices do not drive the findings, the analysis reports a sensitivity sample that excludes the constructed tail.

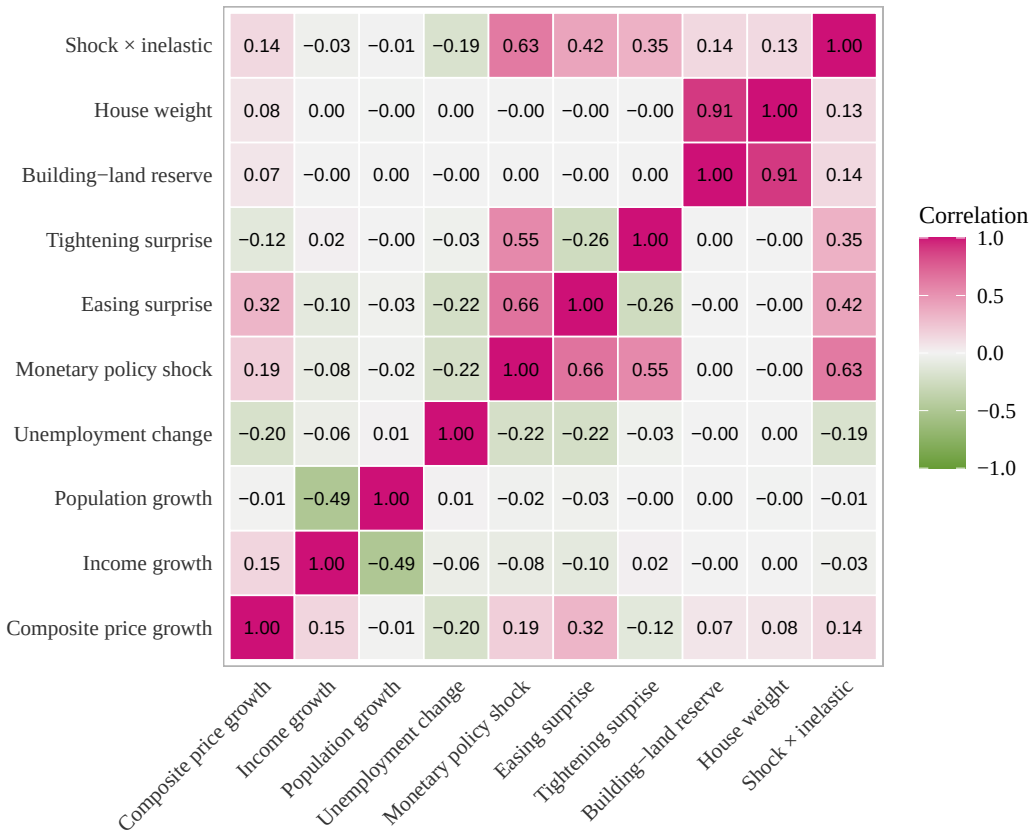


Figure 4: Pairwise correlations of key variables
Pooled Pearson correlations over all district-quarter observations, computed on pairwise-complete cases.

Figure 4 reports the pairwise correlations among the outcome, the controls, the shock and its signed components, the supply measures, and the shock-by-inelastic interaction. The matrix serves two purposes, it documents that the shock behaves as an external regressor with respect to district characteristics, and it surfaces one strong association that shapes the interpretation of the results.

The most reassuring result for identification is the relation of the shock to the district-level variables. The shock is essentially uncorrelated with the two time-invariant supply measures, with a coefficient at 0.00 with both the building land reserve and the fixed house weight. Further, it correlates only weakly with the district controls, at −0.08 with

income growth, −0.02 with population growth, and −0.22 with the change in unemployment. The near-zero correlation with the time-invariant measures is mechanical rather than coincidental, since a common national surprise carries no cross-sectional variation that could line up with fixed district traits, but it nonetheless confirms that the identifying variation in the shock does not proxy for the supply classification. The modest negative correlation with the unemployment change is consistent with monetary easing coinciding with weaker labor market conditions, and it motivates the inclusion of the controls rather than raising a collinearity concern.

The shock also relates to the outcome in the expected direction. The net shock correlates at 0.19 with quarterly composite price growth, where the signed decomposition is more informative, with the easing component correlating at 0.32 and the tightening component at −0.12. Under the convention that easing surprises enter negatively, these signs indicate that price growth is higher in quarters with larger easing surprises. Within the shock block, the net measure correlates at 0.66 with the easing component and at 0.55 with the tightening component, while the two signed components correlate at −0.26 with each other, reflecting the within-quarter cancellation documented in Section 4.2. The interaction term correlates at 0.63 with the net shock, as expected since it is the shock scaled by the relative share of districts classified as inelastic, and only weakly, at around 0.13 to 0.14, with the two supply measures.

Among the controls, the strongest association is the −0.49 correlation between income growth and population growth. This is largely mechanical, as income per capita is constructed by dividing an annual district income total by quarterly population, so that movements in the denominator enter both series with opposite sign, as noted in Section 4.4. The remaining control correlations are small, with composite price growth correlating at −0.20 with the change in unemployment and at 0.15 with income growth, which is consistent with local demand conditions without indicating redundancy among the regressors.

The one association that requires closer attention is the 0.91 correlation between the building land reserve and the fixed house weight. Because inelastic districts are predominantly urban and apartment-heavy, the supply classification and the composite property mix carry closely related cross-sectional information, so that the inelastic indicator partly captures the difference between apartment markets and house markets rather than land scarcity alone. This entanglement is the central measurement concern

of the analysis.

By construction, apartments carry more weight in the housing stock of urban and therefore inelastic districts. Simultaneously, apartment-heavy markets are more strongly shaped by investor demand than owner-occupied house markets. This matters because the two segments do not respond to monetary policy in the same way. Biljanovska et al. (2025) document that investor demand, measured through non-owner-occupied purchases, contracts more sharply after monetary tightening, a pattern they attribute to investors extending recent price trends into their expectations of future returns, which makes their demand particularly sensitive to financing conditions. Investors evaluate housing as one asset among others and discount expected future returns at market interest rates, so that a shift in the policy stance revalues investor-held property more forcefully than owner-occupied housing, where the decision to buy or sell is tied to housing needs rather than to the rate of return (He & La Cava, 2020). The behavioral evidence points in the same direction, as Genesove and Mayer (2001) find that owner-occupants adjust their asking prices far more reluctantly than investors when facing nominal losses, so that owner-occupied house markets are the stickier of the two segments. The Austrian tenure structure is consistent with this reasoning. Using the 2024 building and housing census and the same building-size classification that defines the composite weights, the stock-weighted owner-occupancy rate is only about 19% in Vienna, whose districts are all classified as inelastic, against about 61% in Austria excluding Vienna (Statistik Austria, 2026c). The supply-inelastic set is therefore also the part of the country where rental and investor-held dwellings dominate, precisely the segment the literature identifies as most sensitive to monetary policy.

This literature therefore indicates that the apartment-heavy composition of inelastic districts is not merely a statistical nuisance but a channel that plausibly reinforces the supply mechanism, since the districts classified as inelastic are also those in which the more monetary-sensitive investor segment is most prominent. The two channels nonetheless cannot be fully separated within the current design, as the inelastic indicator jointly reflects land scarcity and property composition. The analysis therefore does not claim that the interaction isolates a pure supply-elasticity effect, and it addresses the concern directly through the property-specific outcomes and the continuous reserve measure reported as alternative specifications, together with the Vienna-excluded robustness check, which the empirical strategy addresses in detail.

4.5. Panel Construction and Preliminary Diagnostics

The estimation panel is assembled by merging the price series, the composite weights, the supply-constraint classification, the quarterly shock, and the three control variables on the district and quarter keys. The price series and controls vary by district and quarter, the shock is common across districts within a quarter, and the supply classification and composite weights are time-invariant district characteristics. Districts are identified by a consistent numeric code across all sources, and the Viennese municipal districts are aligned to the three-digit district scheme used throughout the panel.

A set of preliminary diagnostics confirms that the merge preserves the intended structure of the data. All component inputs join without duplicate or unmatched keys, every district appears in every quarter, and the price series contain no non-positive or non-finite values, so the merged panel is balanced and complete at 4 176 district-quarter observations (covering 116 districts over 36 quarters). Summary statistics for the outcome, the shock, the supply measure, and the controls, together with the integrity checks underlying these statements, are reported in Appendix 3. The descriptive evidence discussed above draws on this panel, and the diagnostics reported here establish that the regularities documented in the preceding subsections reflect the data.

The dependent variable and the lagged controls introduce structural missing values by construction rather than through gaps in the source data. The cumulative outcome at horizon h, the change in the log composite price measured relative to the quarter preceding the shock, together with the four lagged quarterly changes that enter as controls, reduces the usable sample from 3 596 observations at the shortest horizon to 2 204 at the longest. All 116 districts remain represented at every horizon, but each additional horizon removes the most recent quarter, so that comparisons of coefficients across horizons also compare progressively earlier samples. This feature is inherent to the local projection design and is disclosed alongside the estimates.

5. Empirical Strategy

The following chapter describes how the district-quarter panel is used to estimate how the dynamic response of real estate prices to ECB monetary policy shocks differs between supply-elastic and supply-inelastic districts. The approach is a panel local projections

analysis, which traces the price-response over a three-year horizon and identifies the differential effect through the interaction of the shock with the supply-constraint classification. The following subsections introduce the local projection method, specify the baseline model and its identifying assumptions, explain the interpretation of the coefficient of interest, describe the outcome transformation and the fixed-effects structure, and discuss the treatment of inference. Further, the concluding subsection in this chapter discusses alternative model specifications and robustness checks.

5.1. Panel Local Projections

The question motivating this thesis is inherently dynamic, since the interest lies not only in whether prices react to a monetary policy shock but in how that reaction unfolds over the quarters that follow. A common way to trace such a reaction is vector autoregression, which summarizes the joint dynamics of a set of variables in a single system and then projects that system forward to obtain the response at each horizon. The drawback is that the response at a distant horizon is built from the same estimated system used at the nearest one, so that any error in how the system is specified accumulates as the horizon grows (Jordà, 2005). For a study concerned with responses up to three years ahead, this accumulation is a concern.

Local projections offer a more direct route and are the method adopted here. Rather than relying on a single system, a separate regression is run for each horizon, relating the price outcome several quarters ahead to the shock and to controls measured at the time of the shock, and the response is obtained from the coefficients on the shock across this sequence of regressions (Jordà, 2005). The appeal of this method for the setting of this thesis is threefold. It is estimated by ordinary least squares (OLS) and thus extends without difficulty to a panel with district and time fixed effects. Because each horizon is estimated separately, a misspecification of the dynamics at one horizon does not contaminate the others. And it accommodates the interaction between the shock and a district characteristic that lies at the center of this analysis, which would be far less straightforward to impose within a full multivariate system. One drawback is that the outcomes used across horizons overlap in time, which leaves the residuals serially correlated and requires care in the treatment of standard errors, a concern discussed in detail below.

These properties carry over naturally to a panel, for which Jordà (2023) describes local projections as a way of connecting the impulse-response tradition of macroeconomics with the treatment-effect reasoning of applied microeconomics, with the shock playing the role of an intervention whose effect is followed through time. In this thesis the panel does more than accommodate the method. Since the shock is the same for every district in a given quarter, its overall effect on Austrian real estate prices cannot be separated from anything else happening nationally in that quarter, and it is not the object of interest. What can be identified is how districts differ in their reaction according to their supply conditions, which follows from interacting the common shock with the district-level classification. The construction of that interaction, and the fixed-effects structure that isolates the differential it captures, are developed in the following subsections.

5.2. Baseline Model and Identification

The baseline specification follows the panel local projection design of Aastveit and Anundsen (2018), adapted to the Austrian district-level setting. For each horizon $h = 0, 1, \ldots, 12$, a separate regression is estimated of the form:

$$y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \tau_{t,h} + \beta_h \left(\text{Shock}_t \times \text{Inelastic}_i\right) + \theta_h' X_{i,t} + \varepsilon_{i,t+h}, \tag{5.1}$$

where i indexes districts and t quarters. The dependent variable $y_{i,t+h} - y_{i,t-1}$ is the cumulative change in the log composite price between the quarter preceding the shock and horizon h, so that the sequence of coefficients traces the response of prices over a maximum period of three years. The outcome construction is described in detail below. The horizon runs from the quarter of the shock, $h = 0$, to twelve quarters ahead, which spans the three-year window over which the price response is of interest.

The regressor of interest is the interaction $\text{Shock}_t \times \text{Inelastic}_i$, formed from the quarterly net OIS 20Y surprise and the time-invariant supply classification. Here $\text{Inelastic}_i \in \{0, 1\}$ is an indicator equal to one for districts classified as supply-inelastic and zero otherwise, constructed as described in Section 4.3. Its coefficient β_h is the object the analysis seeks to estimate, and it measures how the price response at horizon h differs between inelastic and elastic districts. The binary indicator is used as the baseline because it yields a directly interpretable differential between elasticity-groups. The same supply

constraint is also measured continuously, through the underlying building land reserve, and the analysis reports that specification alongside the binary one as a complementary and more efficient use of the supply information; the two are developed together in the concluding subsection. The term $\alpha_{i,h}$ denotes district fixed effects and $\tau_{t,h}$ quarter fixed effects, both estimated separately at each horizon; their role in isolating the differential response is discussed below. The vector $X_{i,t}$ collects the control variables, entered as four quarterly lags each of the change in the log composite price, the change in log income per capita, population growth, and the change in the unemployment rate, following the lag structure motivated in Section 4.1.

Two features of this specification deserve emphasis. First, the shock enters only through its interaction with the supply classification, not on its own. Because the surprise is common to all districts within a quarter, its level is perfectly collinear with the quarter fixed effects and cannot be separately estimated, so the specification is designed to recover the differential response rather than the aggregate one. Second, the coefficient of interest is allowed to vary freely across horizons, since each β_h is estimated in its own regression, which is what permits the response to build, peak, and fade over the three-year window without imposing a particular dynamic shape. The interpretation of β_h, including the role of the sign convention on the shock, is developed below, as are the assumptions required for it to carry a causal reading.

5.3. Outcome Transformation

The dependent variable is a cumulative log-price change, which is the standard outcome in the local projection literature (Jordà, 2005, 2023) and the one used by Aastveit and Anundsen (2018). Let $p_{i,t}^{\text{comp}}$ denote the composite price of district i in quarter t, constructed as described in Section 4.1, and let $y_{i,t} = \ln p_{i,t}^{\text{comp}}$ be its natural logarithm. The outcome at horizon h is the change in this log price between the quarter preceding the shock and h quarters after it,

$$y_{i,t+h} - y_{i,t-1} = \ln p_{i,t+h}^{\text{comp}} - \ln p_{i,t-1}^{\text{comp}}, \tag{5.2}$$

so that a separate outcome is formed for each horizon $h = 0, 1, \ldots, 12$. Because it is a difference in logarithms, the outcome measures the cumulative growth of the composite

price from the pre-shock quarter to horizon h, and for the moderate changes observed over the sample it is closely approximated by the corresponding percentage change. Expressing the response as a proportional change also places every district on a common percentage scale, so that a euro movement counts relative to the local price level rather than in absolute terms. This matters in a panel where the composite price ranges from under € 900 to over € 13 000 per m^2, since it prevents the response from being mechanically larger in the most expensive districts. The district fixed effects remove differences in the price *level*, whereas the log transformation governs the scale on which the change itself is measured.

Two aspects of this construction shape the interpretation of the estimates. First, the outcome is measured relative to quarter $t-1$ rather than quarter t. Anchoring the base in the quarter before the shock ensures that the horizon-zero outcome, $y_{i,t} - y_{i,t-1}$, already captures the immediate within-quarter movement in prices, so that the response is measured from a point that is not itself contaminated by the shock. Second, the outcome is cumulative rather than a period-by-period change. Each β_h therefore describes the differential response of the price *level* by horizon h, measured from its pre-shock value, rather than the differential growth in that single quarter. The sequence $\{\beta_h\}_{h=0}^{12}$ consequently traces the full path of the accumulated price response over the three-year window, which is the object of the temporal sub-question introduced in Section 1.

Forming the outcome from leads of the log price mechanically removes the final quarters of each district from the longer-horizon regressions, since the price twelve quarters ahead is not observed for the most recent quarters of the sample. The resulting decline in the estimation sample with the horizon, and its implication that longer-horizon coefficients rest on progressively earlier quarters, is documented in Section 4.5 and is revisited when the results are interpreted.

5.4. Interpretation of the Coefficient

The coefficient β_h is the central estimate of the analysis, and its interpretation follows directly from the structure of the model. Since the shock enters only through its interaction with the inelastic indicator, β_h measures the difference in the cumulative price response at horizon h between a supply-inelastic and a supply-elastic district exposed to the same monetary policy surprise. In the language of dynamic treatment effects, to

which Jordà (2023) connects the local projection, β_h is the additional effect of the shock on inelastic districts relative to the elastic benchmark, traced out over the three-year window. It is not the response of any district in isolation, and it is not the average response across all districts, both of which are bound up with the common time effect and cannot be recovered from this specification.

The sign of β_h must be read together with the sign convention on the shock, and the two combine in a way that is easily misread. A negative OIS 20Y surprise denotes monetary easing (see Section 4.2), so the shock variable takes negative values in easing quarters. The differential effect on the cumulative price response of an inelastic district is the product $\beta_h \times \text{Shock}_t$. If, as the mechanism in this thesis suggests, easing raises prices more strongly where supply is constrained, then this product must be positive in an easing quarter, which requires β_h to be negative. A negative estimate of β_h therefore supports the hypothesis of a stronger response in inelastic districts. To avoid this ambiguity, the results are reported and discussed in terms of the implied response to a monetary easing surprise of a given size, rather than in terms of the raw coefficient alone.

This differential reading is the direct empirical counterpart of the research question, and it mirrors the approach of Aastveit and Anundsen (2018), who interpret the interaction between a monetary policy shock and a measure of supply elasticity as the extent to which tighter supply amplifies the price response. The magnitude of β_h is expressed in the units of the outcome, so that it represents the differential cumulative change in the log composite price, in log points, by horizon h, in response to a one-unit surprise. The full sequence $\{\beta_h\}$ describes how this differential builds and evolves over the twelve quarters, which speaks to the temporal sub-question, while its sign and persistence speak to whether supply constraints amplify or merely delay the transmission of monetary policy to regional prices. Whether β_h can be given a causal rather than a purely associational reading depends on the identifying assumptions discussed below.

5.5. Fixed Effects

The specification includes both district and quarter fixed effects, estimated separately at each horizon, and each serves a distinct purpose. District fixed effects $\alpha_{i,h}$ absorb all time-invariant characteristics of a district, including its average price level, its position

in the supply-constraint classification, and its fixed property composition. This matters given the concern raised in Section 4.4 that the inelastic indicator is closely tied to an apartment-heavy housing stock. By removing the cross-sectional level differences between districts, the district effects ensure that β_h is not driven by the fact that inelastic districts are more expensive or more urban on average, but by how their prices move differently over time.

Quarter fixed effects $\tau_{t,h}$ are what allow the model to be read as a differential response. They absorb everything common to all districts within a given quarter, including the level of the monetary policy shock itself, the national housing cycle, euro-area financial conditions, and any other aggregate development. Because the shock does not vary across districts within a quarter, its level is perfectly collinear with the quarter fixed effects and drops out of the regression, as noted in Section 5.2. The only variation left to identify β_h is therefore cross-sectional and within-quarter, namely the difference in how inelastic and elastic districts respond to the same aggregate surprise. This is a deliberate feature rather than a limitation, since it removes at a stroke the large class of confounders that operate at the national level and would otherwise contaminate an estimate of the aggregate response.

This two-way structure follows Aastveit and Anundsen (2018), who combine regional fixed effects with the interaction between the shock and supply elasticity, and whose extended specification adds region-by-time effects to absorb common regional developments. In the present setting the quarter fixed effects already absorb all national variation, and the panel is not large enough along the cross-sectional dimension to add Bundesland-by-quarter fixed effects, the Austrian analogue of these region-by-time effects, without exhausting the identifying variation. The fixed effects carry the horizon subscript h because a separate regression is estimated at each horizon, so that the absorbed level differences and common shocks are allowed to matter differently at different points of the response path. In practice each horizon regression is estimated by ordinary least squares (OLS) with the two sets of fixed effects absorbed through the within transformation, which removes the district and quarter means without estimating a separate coefficient for each unit. What this structure cannot absorb are district-specific developments that are correlated with the supply classification and that happen to coincide with the timing of the shocks, and the plausibility of ruling can be found below.

5.6. Inference and Identification Assumptions

Valid inference is the most delicate part of the design, because the residuals of a panel local projection are dependent along two dimensions at once. The first dimension is temporal. Because the outcome at each horizon accumulates the price change from the same base quarter, the regressions at adjacent horizons are built from overlapping windows of data, so their residuals are expected to be serially correlated within districts. Subsequently, residuals are expected to display stronger serial correlation at longer lags, where the accumulation windows overlap for more quarters. The price momentum documented in Section 4.1 reinforces this concern, since persistent local growth is itself a source of within-district serial correlation. The second dimension is cross-sectional. Districts are exposed to the same national shock and the same aggregate housing cycle, and neighboring markets are linked through common regional developments, so the residuals are expected to be correlated across districts within a given quarter. Ordinary standard errors, which assume independence, would understate the true sampling uncertainty on both counts and cannot be used.

The inference reported throughout rests on the standard errors of Driscoll and Kraay (1998), which extend the nonparametric time-series correction of Newey and West (1987) to a panel and remain valid under broad patterns of dependence both across districts and over time. This estimator is the natural choice in the present setting for two reasons. It addresses the serial and the cross-sectional dependence together, matching the two-dimensional structure of the residuals, and it is the correction used in the regional local projection literature on which this thesis draws, most directly Aastveit and Anundsen (2018). An alternative would be two-way clustering by district and quarter in the manner of Cameron et al. (2011), but its asymptotics require many clusters in both dimensions, and with only 36 quarterly clusters the time dimension would be the binding constraint, so the clustered errors would tend to be understated. The Driscoll-Kraay standard errors are therefore preferred.

The Driscoll and Kraay (1998) asymptotic justification nonetheless relies on a growing time dimension, which is a genuine limitation at 36 quarters, and this is the point at which the overlapping-horizon serial correlation is most likely to distort inference. To confront this directly, the analysis reports the lag-augmentation approach of Montiel Olea and Plagborg-Møller (2021) as a robustness check on inference. Adding lags of

the dependent variable to each horizon regression absorbs the serial correlation that the projection structure induces, so that heteroskedasticity-robust standard errors become valid without relying on a large time dimension, which is precisely the assumption under strain here. Treating this more recent, projection-specific method as a check on the Driscoll-Kraay results guards against the small-sample weakness of the primary estimator without making statistical significance conditional on the agreement of several corrections, which would raise the risk of discarding genuine effects.

The corrections just discussed concern the precision of the estimates, that is, whether the standard errors around β_h are trustworthy. Reading β_h as a causal differential rather than a mere correlation is a separate matter, and it rests on three identifying assumptions that the fixed effects and the standard errors cannot by themselves secure.

(A1) Shock exogeneity. The monetary policy surprise must be exogenous with respect to Austrian district housing conditions. This is the rationale for the high-frequency identification of Altavilla et al. (2019), which isolates the component of policy that is unanticipated by measuring asset-price movements in a tight window around ECB announcements, so that the surprise is plausibly uncorrelated with developments already underway in regional housing markets. The main threat to this assumption is the central-bank information effect, whereby announcements also convey the central bank's assessment of the economic outlook (Jarociński & Karadi, 2020). Since the shock enters only in interaction with a district characteristic, however, any such aggregate information content is absorbed by the quarter fixed effects and does not by itself bias the differential.

(A2) No confounding district characteristic. No omitted district characteristic may be both correlated with the supply classification and differentially responsive to the common shock. This is precisely the class of confounder that the fixed effects cannot absorb, as noted in Section 5.5, and it is the reason the entanglement between supply inelasticity and property composition documented in Section 4.4 matters for interpretation rather than only for description. The analysis does not claim to rule this out by assumption, but confronts it through the alternative specifications and robustness checks described in Section 5.7, which measure the supply constraint continuously as well as through the binary split, isolate the house and apartment segments where each is economically material, and re-estimate the model on the sample excluding Vienna.

(A3) Validity of the supply proxy. The building land reserve must be a valid proxy for the elasticity of local housing supply. As discussed in Section 4.3, the reserve is a predetermined measure of the scope for additional construction rather than a directly estimated elasticity. Taken together, assumptions (A1) to (A3) define the conditions under which β_h can be read as the causal differential response of supply-inelastic districts, and the discussion of results returns to their plausibility where relevant.

5.7. Alternative Specifications and Robustness

The analysis reports a set of departures from the baseline specification that fall into two groups, distinguished by their purpose. Alternative specifications change the model of the effect itself and are reported alongside the baseline, because they characterize the response rather than merely testing its durability. Robustness checks instead vary an estimation or sample choice while leaving the effect of interest unchanged, in order to show that the baseline finding is not an artifact of a particular decision. The distinction follows the practice of Aastveit and Anundsen (2018), and each of these departures is motivated by a specific feature of the data documented in Section 4.

Four alternative specifications are considered. **(AS1) Property-specific outcomes.** The model is re-estimated with the house and apartment series as separate outcomes in place of the composite, which tests whether the composite result is an artifact of the fixed property-mix weighting and speaks to the entanglement between supply inelasticity and property composition discussed in Section 4.4. In its unweighted form this split treats every district equally, regardless of whether a given segment is economically material there, so the apartment response is estimated in part on districts whose stock is overwhelmingly houses, and conversely. **(AS1b) Stock-weighted property-specific outcomes.** To address this, the property-specific regressions are re-estimated by weighted least squares, weighting each district-quarter by the fixed stock share of the segment being modeled, the house share (w_i^{house}) for the house outcome and the apartment share (w_i^{apt}) for the apartment outcome. A district then contributes to a segment in proportion to how relevant that segment is to its local market, so thinly represented segments add little noise, which is the economically appropriate way to isolate a property-type response given the heterogeneous composition documented in Section 4.1. Because weighting toward apartment-heavy districts would compress the

variation of the binary indicator, the weighted specification uses the continuous reserve interaction introduced below. **(AS2) Asymmetric shock decomposition.** The net shock is decomposed into its easing and tightening components, which is required to address the sub-question on whether the transmission of monetary policy is asymmetric across the phases of the cycle. The decomposition is estimated on both the binary and the continuous supply measure, so that the asymmetry can be read through the same two lenses as the level response. For each horizon h, this specification replaces the single interaction in Equation 5.1 with two,

$$\begin{aligned} y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \tau_{t,h} + \beta_h^{\text{ease}} \left(\text{Shock}_t^{\text{ease}} \times \text{Inelastic}_i\right) \\ + \beta_h^{\text{tight}} \left(\text{Shock}_t^{\text{tight}} \times \text{Inelastic}_i\right) + \theta_h' X_{i,t} + \varepsilon_{i,t+h}, \end{aligned} \tag{5.3}$$

where $\text{Shock}_t^{\text{ease}}$ and $\text{Shock}_t^{\text{tight}}$ are the within-quarter sums of the negative and positive OIS 20Y surprises respectively, so that β_h^{ease} and β_h^{tight} measure the differential response of inelastic districts to easing and tightening surprises separately. **(AS3) Continuous supply measure.** The binary inelastic indicator is replaced with the building land reserve entered continuously, standardized so that a higher value denotes a more supply-constrained district and its coefficient is comparable in direction to the binary interaction. Although listed here among the alternative specifications, this measure is reported alongside the baseline rather than as a mere durability check, for an econometric reason. Dichotomizing a continuous characteristic into two groups discards the gradient of supply conditions and identifies the differential only from the between-group contrast, which is a known source of lost information and reduces statistical power relative to using the variable in its continuous form. The continuous measure retains the full cross-sectional variation in the reserve, so that, under the maintained assumptions, it estimates the differential more efficiently than the binary split of the same information. Reporting the two together lets the binary specification serve as the readily interpretable anchor, an inelastic-versus-elastic contrast, while the continuous specification shows whether the same pattern holds once the coarse split is relaxed. They are two complementary specifications of a single mechanism, not a headline result and its robustness check.

Five robustness checks probe the durability of the baseline. **(R1) Vienna excluded.**

The model is re-estimated on the sample excluding the Viennese districts, since all of Vienna falls into the inelastic group and its apartment-dominated markets are structurally distinctive, so this check confirms that the differential is not driven by Vienna alone. **(R2) Pre-tail sample.** The model is re-estimated on the sample that excludes the constructed tail of the control series, where income is extrapolated and unemployment and population are carried forward, as documented in Section 4.4, to confirm that the imputed observations do not drive the findings. **(R3) Alternative shock instruments.** The twenty-year and thirty-year German government bond yield surprises are used in place of the OIS 20Y instrument, which checks that the result does not hinge on the particular measure of the shock. **(R4) Alternative classification threshold.** The building-land threshold that separates elastic from inelastic districts is varied, which complements the continuous-measure specification in guarding against an arbitrary cut-off. **(R5) Lag-augmented inference.** The lag-augmentation approach of Montiel Olea and Plagborg-Møller (2021), described in Section 5.6, is applied as a check on the Driscoll-Kraay standard errors that carry the reported inference, since its validity does not rely on the growing time dimension that the primary estimator assumes. This guards against the small-sample weakness of the primary estimator at the length of the panel, rather than making significance conditional on the agreement of several corrections.

6. Empirical Results

This chapter reports the estimated differential response of district real estate prices to ECB monetary policy surprises. It proceeds from the baseline specification, presented together with the continuous supply measure specification, to the alternative specifications that characterize the response, and concludes with the robustness checks that examine its durability. The discussion here is confined to the empirical results of the estimates, their sign, magnitude, precision, and shape across the horizon. The economic interpretation of results is taken up in Section 7.

Two conventions govern how the estimates are reported throughout. The first concerns the scale of the shock. A raw coefficient (β_h) is the response to a one-unit monetary surprise, that is, a monetary surprise of one basis point. The coefficients are therefore re-scaled, and are read as the response to a shock of one standard deviation of the net

quarterly surprise, measured across the 36 quarters, and stated in percent. The shock is a signed net surprise, negative in easing quarters and positive in tightening quarters. The baseline coefficient measures a symmetric response to a movement in this signed measure. The reported response evaluates this coefficient at a one-standard-deviation easing surprise, that is, it multiplies β_h by the negative of one standard deviation. This makes the sign read intuitively. Under the convention of Section 5.4 the raw coefficient is expected to be negative when the mechanism holds. Multiplying a negative coefficient by a negative easing surprise yields a positive number, so throughout this chapter a positive value denotes a larger price increase in supply-inelastic districts following monetary easing. The separation of easing from tightening surprises is a feature only of the asymmetric specification in Section 6.2. Elsewhere the response is symmetric and the easing framing is a reporting device, not a restriction of the model. In these symmetric specifications a tightening surprise of the same size is estimated to have an equal and opposite effect by construction, so a reported easing response of a given sign implies a tightening response of the same magnitude and the opposite sign.

The second convention concerns the specifications where the supply measure enters continuously. Here, the building land reserve enters in standardized form. The raw building land reserve is demeaned and divided by its standard deviation across the 116 districts, and oriented so that a higher value denotes a more supply-constrained district. Its coefficient is therefore the differential response per one standard deviation of the reserve, whereas the binary coefficient is a difference between two supply-groups of districts. The two are not comparable in magnitude and are compared instead on the sign, shape, and precision of the estimated response.

6.1. Baseline Results

The baseline estimates the differential response of the composite price ($p_{i,t}^{\text{comp}}$) using the binary inelastic classification. The estimated response, shown in the left panel of Figure 5, is small and imprecisely estimated at every horizon. It begins slightly negative at the horizons immediately following the monetary surprise, turns positive from around three quarters ahead, and builds to a peak of about 0.25% near horizon seven before receding over the remainder of the three-year window. The confidence interval, formed from the Driscoll-Kraay standard errors, contains zero at every horizon, so the differential

cannot be distinguished from zero at conventional levels. The point estimates in Table 1 convey the same picture at the tabulated horizons, with the response reaching about 0.14% at horizon ten against a standard error of similar magnitude.

Although the differential is not statistically distinguishable from zero, two features of the point estimates are worth recording. From horizon three onward, the sign is in line with the hypothesized supply mechanism, showing a stronger price response where construction is more constrained. The profile is a delayed, hump shape rather than an immediate jump, in line with the price momentum documented in Section 4.1. The peak differential of about 0.25% is also small relative to the roughly 1.3% average quarterly composite price growth over the sample, so that the supply channel, if present at the estimated magnitude, would add only modestly to the price dynamics that districts experience in any case.

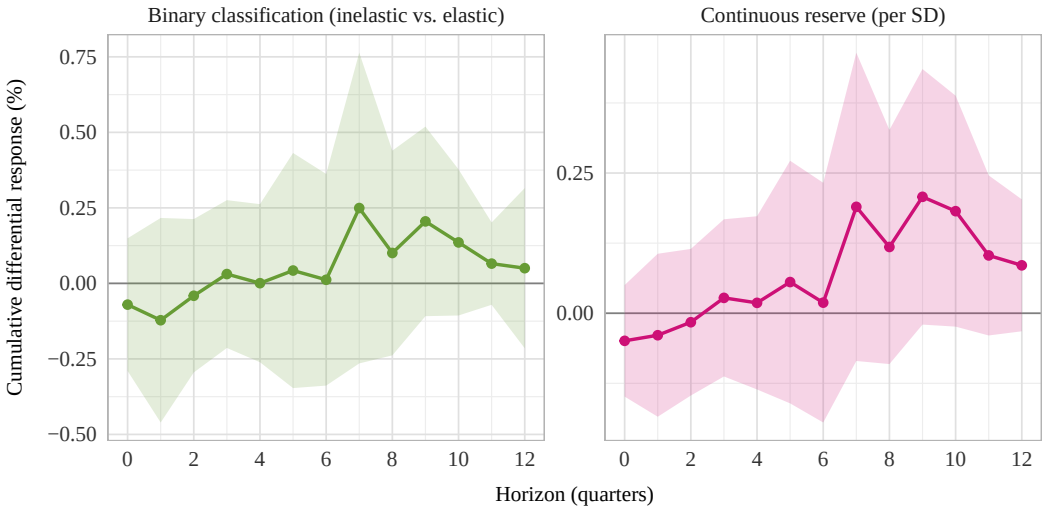


Figure 5: Differential price response, binary and continuous supply measures
Cumulative differential response of the composite price to a one-standard-deviation easing surprise, in percent. Left, binary inelastic classification; right, continuous building land reserve (per standard deviation). Shaded bands are 95% Driscoll-Kraay confidence intervals. The two panels use different units and are compared on shape and precision, not level.

The continuous specification (**AS3**), shown in the right panel of Figure 5, traces a similar profile. Its response is slightly negative at the first horizons, positive from around three quarters ahead, and rising to a peak between horizons seven and nine before receding.

What distinguishes it from the binary specification is its precision, not its shape. The standard errors are roughly half as wide at most horizons, as the columns of Table 1 show, so that the confidence interval narrows around the same path. At the longer horizons the lower bound of the interval approaches zero without crossing it, so that the differential comes close to being distinguishable from zero. This is the pattern the argument in Section 5.7 anticipated, since retaining the full gradient of supply conditions rather than a binary split uses the cross-sectional variation more efficiently and sharpens the estimate of the same underlying response. Read together, the two specifications describe a delayed, positively signed differential of modest size, estimated imprecisely under the binary split and more sharply, though still short of conventional significance, when supply conditions enter continuously. The raw coefficients at every horizon are reported in Appendix Table 6.

	Binary classification			Continuous reserve		
Horizon h	Estimate	SE	95% CI	Estimate	SE	95% CI
0	-0.070	0.112	[-0.290, 0.149]	-0.049	0.051	[-0.149, 0.050]
2	-0.041	0.130	[-0.295, 0.213]	-0.016	0.067	[-0.147, 0.115]
4	0.001	0.133	[-0.261, 0.262]	0.018	0.079	[-0.136, 0.173]
6	0.012	0.179	[-0.338, 0.362]	0.019	0.109	[-0.195, 0.233]
8	0.101	0.173	[-0.238, 0.440]	0.118	0.107	[-0.091, 0.327]
10	0.136	0.123	[-0.106, 0.377]	0.182	0.105	[-0.024, 0.388]
12	0.050	0.135	[-0.215, 0.315]	0.085	0.060	[-0.032, 0.203]

Note. Estimates are the cumulative differential response of the composite price to a one-standard-deviation easing surprise, in percent, with Driscoll-Kraay standard errors and 95% confidence intervals. The binary coefficient is a difference between supply-inelastic and supply-elastic districts; the continuous coefficient is per standard deviation of the building land reserve, and the two are not comparable in magnitude.

Table 1: Baseline differential response, binary and continuous supply measures
Response to a one-standard-deviation easing surprise, in percent, at selected horizons.

6.2. Alternative Specifications

The alternative specifications characterize the response rather than test its durability, and each is reported on the same per-standard-deviation easing scale introduced at the

start of this chapter.

(AS2) Asymmetric shock decomposition. The net shock is split within each quarter into an easing component, the sum of the negative surprises, and a tightening component, the sum of the positive surprises, and the two enter the model with separate interactions. This addresses whether the transmission of monetary policy is asymmetric across the phases of the monetary policy cycle. Figure 6 reports the decomposition under the binary classification. The easing response traces a delayed, positively signed profile that peaks at about 0.28% near horizon seven, close in shape to the baseline, while the tightening response is flatter and less regular around zero. Neither is statistically distinguishable from zero at any horizon under the binary split, so the two responses cannot be distinguished from one another on the basis of these estimates alone, though the point estimates suggest that the easing side is where the differential builds.



Figure 6: Asymmetric differential response, binary supply measure
Cumulative differential response to a one-standard-deviation easing (left) and tightening (right) surprise, in percent, by horizon in quarters, under the binary inelastic classification. Shaded bands are 95% Driscoll-Kraay confidence intervals.

The continuous supply-measure sharpens this picture. Estimating the same decomposition with the building land reserve entered continuously, on the same per-standard-deviation-of-reserve scale as in the baseline, Figure 7 shows an easing response that is distinguishable from zero at the longer horizons, with a differential of about 0.23% at horizon nine (confidence interval $[0.01, 0.46]$) and about 0.14% at horizon twelve (con-

fidence interval $[0.01, 0.27]$), where the interval excludes zero. The tightening response remains centered on zero throughout, with intervals that always contain zero. The easing differential is therefore distinguishable from zero at these horizons while the tightening differential is not, a pattern that is consistent with an asymmetry in which the amplification of the price response in supply-constrained districts operates through monetary easing rather than through tightening. The difference between the two responses is not itself formally tested, so the asymmetry is read as suggestive rather than established. What the estimates show directly is that only the easing side attains a differential distinguishable from zero. This is nonetheless the sharpest evidence in the analysis for the hypothesized channel, and its placement at the longer horizons is consistent with the delayed build seen in the level response. It should be read under the stipulation that these are the horizons at which the estimation sample is smallest and the Driscoll-Kraay lag longest, so the precision is most fragile precisely where the differential clears zero. The raw easing and tightening coefficients for both the binary and the continuous decomposition are reported in Appendix Tables 7 and 8.



Figure 7: Asymmetric differential response, continuous supply measure
Cumulative differential response per standard deviation of the building land reserve to a one-standard-deviation easing (left) and tightening (right) surprise, in percent, by horizon in quarters. Shaded bands are 95% Driscoll-Kraay confidence intervals.

(AS1) Property-specific outcomes. Estimating the house and apartment series as separate outcomes probes whether the composite result reflects a common response

of both segments or is driven by one of them. In its unweighted form, however, this split is uninformative by construction, since it estimates each segment's response with equal weight across all districts. Given the heterogeneity of the district housing stock documented in Section 4.1, where many districts are almost entirely houses or almost entirely apartments, each segment regression is then dominated in part by districts where that segment is a negligible share of the local market and its measured price is thin and noisy. The resulting estimates, reported in Appendix 10, are correspondingly imprecise, with wide intervals and no clear or interpretable profile in either series, and are best read as a motivation for the weighted specification rather than as a result in their own right.

(AS1b) Stock-weighted property-specific outcomes. The weighted specification described in Section 5.7 addresses this by weighting each district-quarter by the fixed stock share of the segment being modeled, so that a district contributes to a segment in proportion to how material that segment is to its local market and thinly represented segments add little noise. This sharpens the property-specific estimates considerably. Figure 8 reports the result, on the continuous building land reserve scale, with the raw coefficients at every horizon in Appendix Table 9. The house response is imprecise throughout but turns mildly positive at the longer horizons, in the direction the mechanism predicts and in contrast to the unweighted split. The apartment response is more pronounced and is distinguishable from zero at horizon six, where the differential is about -0.39% (confidence interval $[-0.62, -0.16]$), before returning toward zero. The negative sign indicates that apartment prices in supply-constrained districts respond less strongly, or in the opposite direction, to easing at this horizon than in less constrained districts, a pattern that runs counter to the house segment and to the composite. The weighted apartment response is estimated predominantly on the urban, apartment-heavy districts, which are also those classified as inelastic and geographically concentrated, as documented in Section 4.3. This segment-level result is therefore bound up with that concentration, whose interpretation is taken up in Section 7.

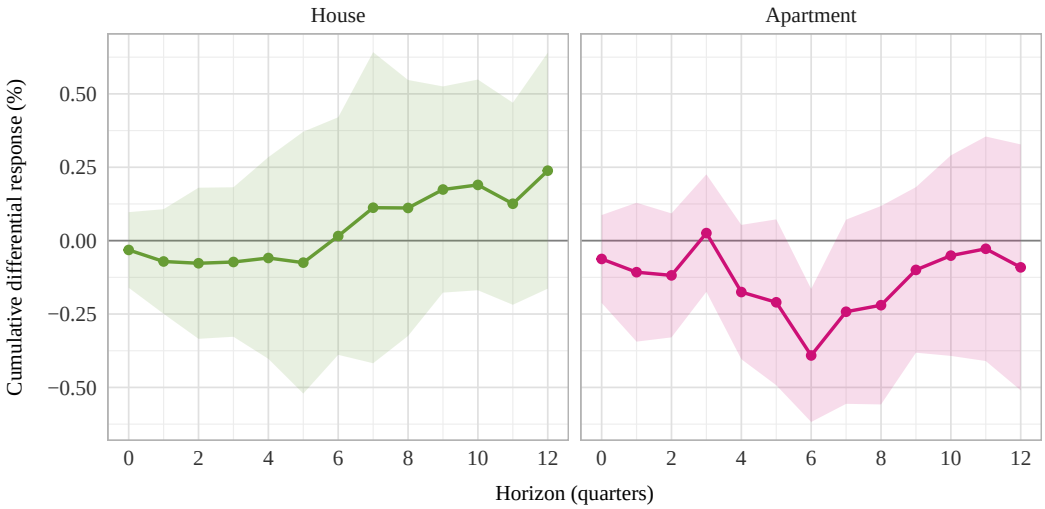


Figure 8: Stock-weighted property-specific differential response
Cumulative differential response of the house (left) and apartment (right) price per standard deviation of the building land reserve to a one-standard-deviation easing surprise, in percent, by horizon in quarters. Each segment is estimated by weighted least squares, weighting district-quarters by that segment's fixed stock share. Shaded bands are 95% Driscoll-Kraay confidence intervals.

6.3. Robustness Checks

Before probing the durability of the findings, the residual diagnostics confirm that the inference underlying the preceding results is appropriate. Estimated on the baseline specification, the within-district first-order autocorrelation of the residuals, the correlation between a district's residual in one quarter and its residual in the next, rises from near zero at horizon zero to about 0.85 at the longest horizon, as the cumulative outcomes at successive horizons come to overlap over increasingly many quarters. The cross-sectional dependence is present as well, with the mean absolute pairwise correlation of the residuals across districts within a quarter rising to about 0.43 at the longest horizon. Both patterns, reported in Appendix 5, confirm the dependence anticipated in Section 5.6 and the need for a correction that addresses both of its dimensions, as the Driscoll-Kraay standard errors used throughout do.

The robustness checks then vary an estimation or sample choice while leaving the effect of interest unchanged, in order to establish that the baseline pattern is not an artifact of a particular decision. Figure 9 overlays the baseline response with the two checks

that bear most directly on the concerns raised in Section 4, the exclusion of Vienna and the use of an alternative shock instrument. The profiles track one another closely through the middle horizons, so the delayed, positively signed shape of the response does not depend on the OIS 20Y instrument or on the presence of Vienna in the sample. The one visible divergence is at the longer horizons, where the Vienna-excluded path flattens and turns slightly negative while the others remain positive, indicating that the magnitude of the differential beyond the two-year horizon is carried disproportionately by the Viennese districts.

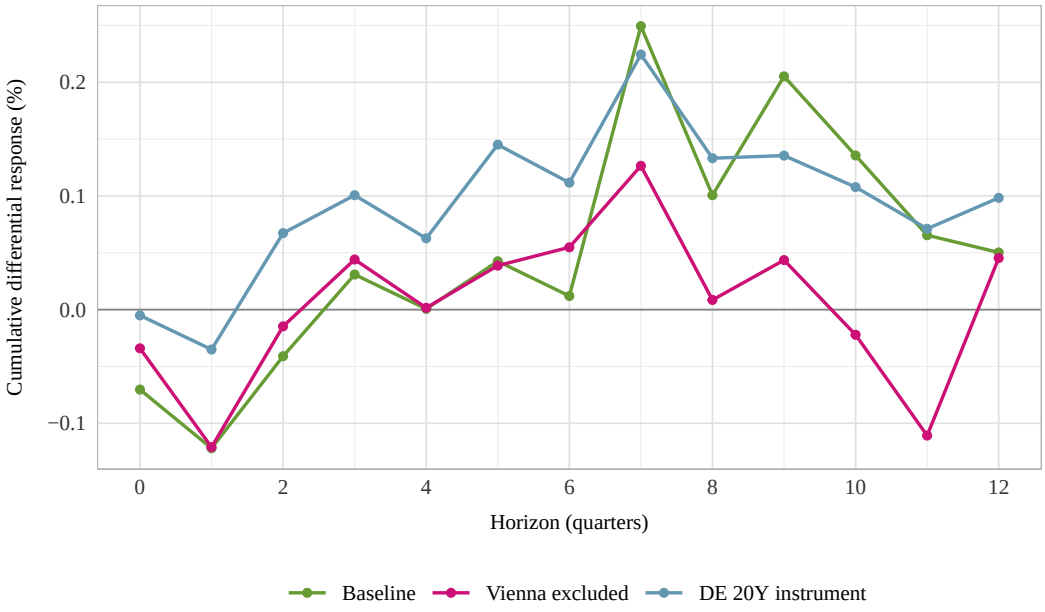


Figure 9: Robustness of the differential response
Cumulative differential response to a one-standard-deviation easing surprise, in percent, by horizon in quarters, for the baseline and two robustness checks. Point estimates only; confidence intervals are omitted for legibility and are reported at focal horizons in Table 2.

Table 2 reports the estimate and confidence interval for every check at the peak horizon, seven quarters ahead, and at the end of the window, twelve quarters ahead. Three patterns are worth noting. First, the peak estimate is stable across the instrument and the classification threshold, ranging from about 0.21% to 0.32% under the German bond instruments and the alternative percentile cut-offs, close to the baseline value of about 0.25%. Second, two checks move the estimate as the data chapter anticipated.

Excluding Vienna roughly halves the peak differential, to about 0.13%, and estimating on the pre-tail sample that drops the extrapolated control observations turns the peak slightly negative, which confirms that the constructed tail contributes to the short-run estimate and motivates its treatment as a sensitivity check. Third, the confidence intervals continue to contain zero at both horizons for every check, with one exception, the lag-augmented estimator, whose heteroskedasticity-robust interval excludes zero at the peak horizon (about 0.25% with interval $[0.01, 0.49]$). Since the lag-augmented estimator shares the baseline point estimate and differs only in the standard error, this indicates that the baseline differential at the peak is distinguishable from zero under an inference procedure whose validity does not rest on the growing time dimension that the Driscoll-Kraay correction assumes, which is reassuring given the short panel, though it remains a single horizon under a single estimator and is not treated as decisive on its own.

	Horizon $h = 7$		Horizon $h = 12$	
Specification	Estimate	95% CI	Estimate	95% CI
Baseline	0.249	[-0.265, 0.764]	0.050	[-0.215, 0.315]
Continuous reserve	0.190	[-0.085, 0.465]	0.085	[-0.032, 0.203]
Vienna excluded	0.127	[-0.430, 0.684]	0.045	[-0.381, 0.472]
Pre-tail sample	-0.027	[-0.490, 0.436]	0.050	[-0.215, 0.315]
DE 20Y instrument	0.224	[-0.032, 0.481]	0.098	[-0.131, 0.328]
DE 30Y instrument	0.212	[-0.099, 0.523]	0.052	[-0.179, 0.283]
Threshold: 20th pct	0.291	[-0.151, 0.732]	0.050	[-0.165, 0.266]
Threshold: 30th pct	0.323	[-0.203, 0.849]	0.146	[-0.165, 0.456]
Lag-augmented	0.249	[0.014, 0.485]	0.050	[-0.270, 0.371]

Note. Each row is a separate estimation. The baseline and the robustness checks use the binary supply classification; the continuous reserve row is included as reference. Intervals are 95% Driscoll-Kraay confidence intervals, except the lag-augmented row, which reports heteroskedasticity-robust intervals following Montiel Olea & Plagborg-Møller (2021).

Table 2: Differential response across robustness checks at focal horizons
Response to a one-standard-deviation easing surprise, in percent, at horizons seven and twelve.

7. Discussion

7.1. Interpretation of Findings

7.2. Relation to Literature

7.3. Economic Relevance

7.4. Policy Implications

7.5. Limitations

8. Conclusion

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A. Appendix

Variable	N	Mean	SD	Min	Median	Max
Outcome						
Composite price (€/m^2)	4 176	3 261	1 593	868	2 876	13 500
House price (€/m^2)	4 176	4 203	2 699	849	3 200	16 212
Apartment price (€/m^2)	4 176	2 865	1 539	742	2 471	13 515
Composite price growth (log change)	4 060	0.0135	0.0238	-0.1080	0.0118	0.1570
Monetary policy shock						
Net shock (OIS 20Y)	36	-0.59	2.66	-5.10	-1.20	5.40
Easing component	36	-2.29	2.30	-9.70	-1.90	0.00
Tightening component	36	1.70	2.06	0.00	0.72	8.50
Supply measure						
Building land reserve (%)	116	21.2	10.2	0.1	23.8	45.5
House weight	116	0.500	0.273	0.003	0.580	0.874
Controls						
Population	4 176	77 446	47 622	1 900	65 370	305 314
Unemployment rate (%)	4 176	6.0	3.3	1.6	5.0	17.9
Net income per capita (€)	4 176	22 680	4 797	5 155	21 680	51 092

Note. N differs by variable block. Prices and controls are observed at the district-quarter level. The monetary policy shock varies only over the 36 quarters. The supply measure and the fixed dwelling-stock weights are time-invariant, they vary over the 116 districts. Composite price growth is a quarterly log change and is undefined in the first quarter of each district, which accounts for its lower count.

Table 3: Summary statistics of the analysis variables

Diagnostic	Result
Balanced panel (districts × quarters)	116 × 36 = 4 176
Unique district-quarter keys	4 176 (no duplicates)
Missing or unexpected keys	0
Districts observed in all 36 quarters	116
Source inputs joined without key errors	7 of 7
Non-positive or non-finite price cells	0
Weight-sum and composite identity errors	0

Table 4: Panel construction and integrity diagnostics

Horizon h	Residual AR(1)	Mean $\lvert\rho\rvert$	Max $\lvert\rho\rvert$
0	-0.01	0.19	0.81
4	0.78	0.30	0.98
8	0.86	0.41	0.99
12	0.85	0.43	0.99

Note. Residuals are from the baseline two-way fixed-effects specification (Equation 5.1), estimated at selected horizons h. Residual AR(1) is the first-order within-district autocorrelation of the residuals; Mean $\lvert\rho\rvert$ and Max $\lvert\rho\rvert$ are the mean and maximum absolute pairwise correlation of the residuals across districts within a quarter. The values rising over horizons reflect the serial and cross-sectional dependence discussed in Section 5.6 and Section 6.3.

Table 5: Residual dependence of the baseline specification

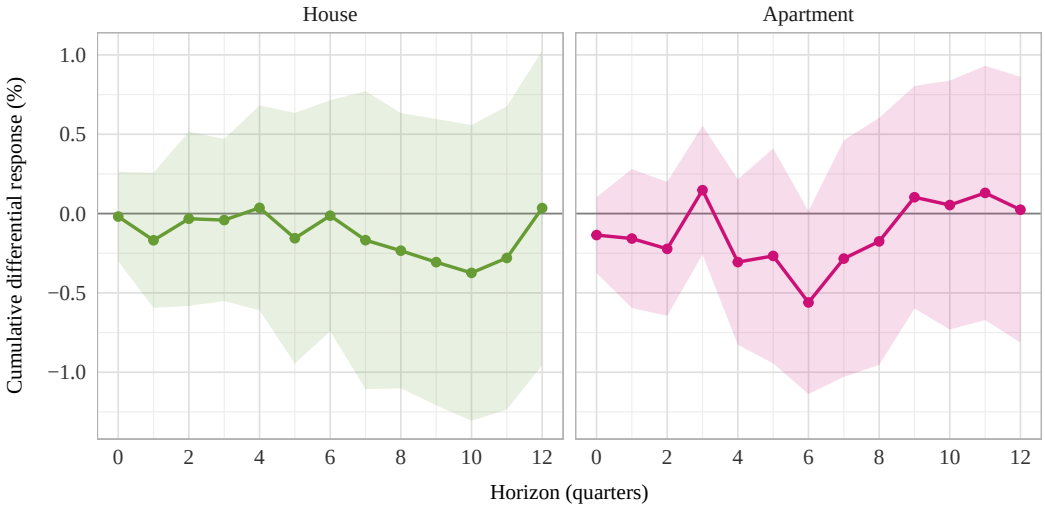


Figure 10: Property-specific differential response, unweighted
Cumulative differential response of the house (left) and apartment (right) price to a one-standard-deviation easing surprise, in percent, by horizon in quarters, under the binary inelastic classification and without stock weighting. Shaded bands are 95% Driscoll-Kraay confidence intervals. This unweighted split is the motivating specification for the stock-weighted estimates in Section 6.2.

	Binary classification			Continuous reserve			
h	β_h	SE	95% CI	β_h	SE	95% CI	N
0	0.0003	0.0004	[-0.0006, 0.0011]	0.0002	0.0002	[-0.0002, 0.0006]	3596
1	0.0005	0.0007	[-0.0008, 0.0017]	0.0001	0.0003	[-0.0004, 0.0007]	3480
2	0.0002	0.0005	[-0.0008, 0.0011]	0.0001	0.0003	[-0.0004, 0.0006]	3364
3	-0.0001	0.0005	[-0.0010, 0.0008]	-0.0001	0.0003	[-0.0006, 0.0004]	3248
4	-0.0000	0.0005	[-0.0010, 0.0010]	-0.0001	0.0003	[-0.0007, 0.0005]	3132
5	-0.0002	0.0007	[-0.0016, 0.0013]	-0.0002	0.0004	[-0.0010, 0.0006]	3016
6	-0.0000	0.0007	[-0.0014, 0.0013]	-0.0001	0.0004	[-0.0009, 0.0007]	2900
7	-0.0009	0.0010	[-0.0029, 0.0010]	-0.0007	0.0005	[-0.0017, 0.0003]	2784
8	-0.0004	0.0007	[-0.0017, 0.0009]	-0.0004	0.0004	[-0.0012, 0.0003]	2668
9	-0.0008	0.0006	[-0.0020, 0.0004]	-0.0008	0.0004	[-0.0016, 0.0001]	2552
10	-0.0005	0.0005	[-0.0014, 0.0004]	-0.0007	0.0004	[-0.0015, 0.0001]	2436
11	-0.0002	0.0003	[-0.0008, 0.0003]	-0.0004	0.0003	[-0.0009, 0.0001]	2320
12	-0.0002	0.0005	[-0.0012, 0.0008]	-0.0003	0.0002	[-0.0008, 0.0001]	2204

Note. Raw coefficients per one-basis-point OIS 20Y surprise, with Driscoll-Kraay standard errors and 95% confidence intervals; N: number of district-quarter observations per horizon. Under the sign convention of Section 5.4 a negative coefficient corresponds to a larger price increase in supply-inelastic districts following monetary easing. Percent response to a one-standard-deviation easing surprise reported in the text: $\beta_h \; x \; -SD \; x \; 100$ (SD = 2.66). The continuous coefficient is per one standard deviation of the building land reserve.

Table 6: Baseline and continuous (AS3) specifications, raw coefficients
Estimated interaction coefficient per horizon h.

	Easing			Tightening			
h	β_h	SE	95% CI	β_h	SE	95% CI	N
0	0.0005	0.0004	[-0.0003, 0.0014]	-0.0001	0.0005	[-0.0012, 0.0010]	3596
1	0.0007	0.0006	[-0.0006, 0.0020]	0.0001	0.0008	[-0.0015, 0.0017]	3480
2	0.0001	0.0006	[-0.0011, 0.0014]	0.0002	0.0007	[-0.0012, 0.0015]	3364
3	-0.0001	0.0006	[-0.0014, 0.0011]	-0.0001	0.0005	[-0.0010, 0.0008]	3248
4	0.0001	0.0006	[-0.0012, 0.0013]	-0.0001	0.0007	[-0.0016, 0.0014]	3132
5	-0.0001	0.0009	[-0.0019, 0.0017]	-0.0002	0.0008	[-0.0019, 0.0014]	3016
6	-0.0001	0.0008	[-0.0017, 0.0014]	0.0001	0.0008	[-0.0014, 0.0016]	2900
7	-0.0011	0.0010	[-0.0031, 0.0009]	-0.0007	0.0011	[-0.0029, 0.0014]	2784
8	-0.0007	0.0007	[-0.0021, 0.0008]	0.0001	0.0007	[-0.0012, 0.0014]	2668
9	-0.0008	0.0006	[-0.0020, 0.0004]	-0.0007	0.0008	[-0.0022, 0.0008]	2552
10	-0.0003	0.0005	[-0.0013, 0.0006]	-0.0008	0.0007	[-0.0021, 0.0005]	2436
11	-0.0001	0.0003	[-0.0007, 0.0005]	-0.0005	0.0007	[-0.0020, 0.0010]	2320
12	-0.0000	0.0005	[-0.0010, 0.0010]	-0.0004	0.0007	[-0.0018, 0.0010]	2204

Note. Raw coefficients per one-basis-point OIS 20Y surprise, with Driscoll-Kraay standard errors and 95% confidence intervals; N is the number of district-quarter observations at each horizon. The easing and tightening components are the within-quarter sums of the negative and positive surprises. Under the sign convention of Section 5.4 a negative easing coefficient corresponds to a larger price increase in supply-inelastic districts following monetary easing.

Table 7: Asymmetric specification (AS2), binary classification, raw coefficients

Estimated easing and tightening interaction coefficients per horizon h.

	Easing			Tightening			
h	β_h	SE	95% CI	β_h	SE	95% CI	N
0	0.0003	0.0002	[-0.0001, 0.0008]	-0.0000	0.0002	[-0.0005, 0.0005]	3596
1	0.0003	0.0003	[-0.0003, 0.0010]	-0.0001	0.0003	[-0.0008, 0.0005]	3480
2	0.0001	0.0004	[-0.0006, 0.0008]	0.0000	0.0004	[-0.0008, 0.0009]	3364
3	-0.0001	0.0004	[-0.0008, 0.0006]	-0.0002	0.0003	[-0.0007, 0.0004]	3248
4	-0.0001	0.0004	[-0.0008, 0.0007]	-0.0001	0.0004	[-0.0009, 0.0008]	3132
5	-0.0002	0.0005	[-0.0013, 0.0008]	-0.0002	0.0005	[-0.0011, 0.0008]	3016
6	-0.0002	0.0005	[-0.0012, 0.0008]	0.0002	0.0005	[-0.0007, 0.0011]	2900
7	-0.0009	0.0006	[-0.0021, 0.0003]	-0.0004	0.0006	[-0.0016, 0.0008]	2784
8	-0.0007	0.0005	[-0.0017, 0.0003]	-0.0001	0.0004	[-0.0008, 0.0007]	2668
9	-0.0009	0.0004	[-0.0017, -0.0000]	-0.0006	0.0005	[-0.0017, 0.0004]	2552
10	-0.0007	0.0004	[-0.0016, 0.0001]	-0.0006	0.0004	[-0.0014, 0.0002]	2436
11	-0.0006	0.0003	[-0.0011, -0.0001]	-0.0000	0.0004	[-0.0008, 0.0007]	2320
12	-0.0005	0.0003	[-0.0010, -0.0000]	-0.0001	0.0003	[-0.0007, 0.0006]	2204

Note. Raw coefficients per one-basis-point OIS 20Y surprise, interacted with the standardized building land reserve, with Driscoll-Kraay standard errors and 95% confidence intervals; N is the number of district-quarter observations at each horizon. Coefficients are per one standard deviation of the reserve. Under the sign convention of Section 5.4 a negative easing coefficient corresponds to a larger price increase in more supply-constrained districts following monetary easing.

Table 8: Asymmetric specification (AS2), continuous reserve, raw coefficients

Estimated easing and tightening interaction coefficients per horizon h.

	House			Apartment			
h	β_h	SE	95% CI	β_h	SE	95% CI	N
0	0.0001	0.0002	[-0.0004, 0.0006]	0.0002	0.0003	[-0.0003, 0.0008]	3596
1	0.0003	0.0003	[-0.0004, 0.0009]	0.0004	0.0005	[-0.0005, 0.0013]	3480
2	0.0003	0.0005	[-0.0007, 0.0013]	0.0004	0.0004	[-0.0003, 0.0012]	3364
3	0.0003	0.0005	[-0.0007, 0.0012]	-0.0001	0.0004	[-0.0008, 0.0007]	3248
4	0.0002	0.0007	[-0.0011, 0.0015]	0.0007	0.0004	[-0.0002, 0.0015]	3132
5	0.0003	0.0009	[-0.0014, 0.0020]	0.0008	0.0005	[-0.0003, 0.0019]	3016
6	-0.0001	0.0008	[-0.0016, 0.0015]	0.0015	0.0004	[0.0006, 0.0023]	2900
7	-0.0004	0.0010	[-0.0024, 0.0016]	0.0009	0.0006	[-0.0003, 0.0021]	2784
8	-0.0004	0.0008	[-0.0021, 0.0012]	0.0008	0.0006	[-0.0004, 0.0021]	2668
9	-0.0007	0.0007	[-0.0020, 0.0007]	0.0004	0.0005	[-0.0007, 0.0014]	2552
10	-0.0007	0.0007	[-0.0021, 0.0006]	0.0002	0.0007	[-0.0011, 0.0015]	2436
11	-0.0005	0.0007	[-0.0018, 0.0008]	0.0001	0.0007	[-0.0013, 0.0015]	2320
12	-0.0009	0.0008	[-0.0024, 0.0006]	0.0003	0.0008	[-0.0012, 0.0019]	2204

Note. Raw coefficients per one-basis-point OIS 20Y surprise, interacted with the standardized building land reserve, with Driscoll-Kraay standard errors and 95% confidence intervals; N is the number of district-quarter observations at each horizon. Each segment is estimated by weighted least squares, weighting district-quarters by that segment's fixed stock share. Coefficients are per one standard deviation of the reserve. Under the sign convention of Section 5.4 a negative coefficient corresponds to a larger price increase in more supply-constrained districts following monetary easing.

Table 9: Stock-weighted property-specific specification (AS1b), raw coefficients

Estimated interaction coefficient for the house and apartment outcomes per horizon h.